Tutorial-03

$h_5 \Rightarrow w_{13} = 0.3,\ w_{14} = 0.3,\ w_{15} = 0.1$

$= 3(0.3) + 5(0.3) + 1(0.1)$

$= 2.5$

$\sigma(z) = \frac{1}{1+e^{-2.5}} = 0.7806$

$0.2(0.880) + 0.1(0.7806) + 0.3(0.880) + 0.2(0.$

$+ 0.4(0.7806) + 0.$

$z = 0.7085$

$\sigma(z) = \frac{1}{1+e^{-0.7085}} = 0.6700$

Training example

$x_1 = 6,\ x_2 = 3,\ x_3 = 2$

$h_1 \Rightarrow w_1 = 0.2,\ w_2 = 0.3,\ w_3 = 0.2$

$= 6(0.2) + 3(0.3) + 2(0.2)$

$z = 2.2$

$\sigma(z) = \frac{1}{1+e^{-2.2}} = 0.9002$

$h_2 \Rightarrow w_4 = 0.1,\ w_5 = 0.4,\ w_6 = 0.2$

$= 6(0.1) + 0.4(3) + 1(0.2)$

$z = 2 \quad \sigma(z) = \frac{1}{1+e^{-2}} = 0.8807$

$h_3 \Rightarrow w_7 = 0.3,\ w_8 = 0.2,\ w_9 = 0.1$

$= 6(0.3) + 0.2(3) + 1(0.1)$

$z = 2.5 \quad \sigma(z) = \frac{1}{1+e^{-2.5}} = 0.7806$

$h_4 \Rightarrow w_{10} = 0.4,\ w_{11} = 0.1,\ w_{12} = 0.2$

$6(0.4) + 0.1(3) + 1(0.2)$

$h_5 = w_{13}$

$= 6(0.3) + 3(0.3) + 0.1(1)$

$z = 2.8 \quad \sigma(z) = \frac{1}{1+e^{-2.8}} = 0.94267$

$0.2(0.9002) + 0.1(0.8807) + 0.3(0.7806) + 0.2(0.9478)$

$+ 0.4(0.9426) + 0.1$

$z = 1.16889$

$\sigma(z) = \frac{1}{1+e^{-1.16889}} = 0.76294$

Training Example-4

$x_1 = 2, \; x_2 = 6, \; x_3 = 1$

$h_1 = w_1 = 0.2, \; w_2 = 0.3, \; w_3 = 0.1$

$= 2(0.2) + 6(0.3) + 1(0.1)$

$z = 2.3 \quad \sigma(z) = \frac{1}{1+e^{-2.3}} = 0.9088$

$h_2 = w_4 = 0.1, \; w_5 = 0.4, \; w_6 = 0.2$

$= 2(0.1) + 0.4(6) + 1(0.2)$

$z = 2.8 \quad \sigma(z) = \frac{1}{1+e^{-2.8}} = 0.9426$

$h_3 = w_7 = 0.3, \; w_8 = 0.2, \; w_9 = 0.1$

$= 2(0.3) + 6(0.2) + 1(0.1)$

$= 1.9$

$z = \frac{1}{1+e^{-1.9}} = 0.8694$

$h_4 = w_{10} = 0.3, \; w_{11} = 0.3, \; w_{13} = 0.1$

$2(0.3) + 6(0.3) + 1(0.1)$

$z = 2.5 \quad \sigma(z) = \frac{1}{1+e^{-2.5}} = 0.924$

$\Rightarrow 0.2(0.988) + 0.1(0.942) + 0.3(0.869) + 0.2(0.83)$

$+ 0.4(0.924) + 0.1$

$z = 1.73 \quad \sigma(z) = \frac{1}{1+e^{-1.73}} = 0.7636$